

Geometry

Unit 8

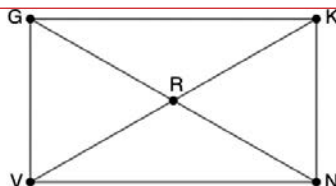
Lesson 3 Homework

Name _____

Date _____

Period _____

Rectangle $GKNV$ is shown with diagonals $\overline{GN}$ and $\overline{KV}$ intersecting at R where $GK = 26x - 13$ and $NV = 14x + 5$.



Commented [DB1]: <https://www.geogebra.org/calculator/zf6n6tqk>

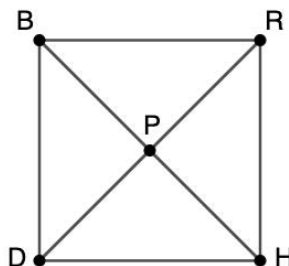
1. Select all true properties of $GKNV$.
 - ☐ Opposite sides are parallel.
 - ☐ Opposite sides are congruent.
 - ☐ Consecutive angles are congruent.
 - ☐ All angles are right angles.
 - ☐ Diagonals are perpendicular.
 - ☐ All sides are congruent.

2. What is the value of x ?

3. What is the length of $\overline{NV}$?

Square $BRHD$ with diagonals $\overline{BH}$ and $\overline{RD}$ intersecting at point P .

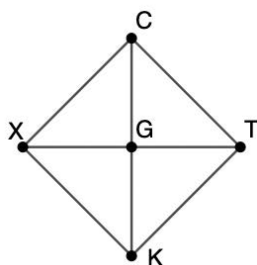
4. Write two congruent statements using the diagonals of $BRHD$.



5. Given $BP = 32y - 10$, and $HP = 18y + 32$. What is the value of y ?

6. What is the length of $\overline{RD}$?

Quadrilateral $CTKX$ is shown with diagonals $\overline{CK}$ and $\overline{XT}$ intersecting at point G .



7. What value of g makes $CTKX$ a rhombus, if $CT = 55$, and $TK = 15g + 19$?

8. Given that $CTKX$ is a rhombus with $CG = 6y - 17.81$ and $KG = 3y + 10.54$. What is the length of $\overline{TX}$?

9. What properties are the same between $GKNV$ and $CTKX$?

10. What properties are different between $BRHD$ and $CTKX$?